

IoT based Predictive Maintenance in Manufacturing Sector

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Abstract: Industry 4.0 enables technological trends like Big Data Analytics and Machine Learning techniques to converge into and merge with traditional manufacturing processes, resulting in smart manufacturing. Smart manufacturing techniques leverage the use of Industrial Internet of things (IIoT) technology using IoT sensors that are fitted on physical assets to enhance manufacturing processes. IIoT Sensors enable smart manufacturing facilities capable of autonomously exchanging information, which can be used to drive business decisions more accurately. Businesses that adopt Smart manufacturing techniques lead to a competitive advantage for these firms as they can bring in higher profit margins, reduced maintenance costs, energy savings, and better-quality products. This study proposes an architecture for IIoT based predictive maintenance. A case study from ancillary automobile industry is presented to demonstrate a predictive model for predicting sudden breakdown in industrial machines, thereby enabling the production and maintenance cycle to be 'smart.'

Keywords: Industry 4.0, Analytic Models, Machine learning, Smart Manufacturing, Sensors

1. Introduction

In today's highly competitive business environment, organizations want to increase production and operational efficiency, employee safety and drive financial results. Many manufacturing organizations are yet to leverage the use of big data and machine learning analytics due to lack of smart analytic tools and capabilities (Jay Lee, Kao, & Yang, 2014). The transition and advancements in manufacturing technologies have defined the industrial revolution. Industry 1.0 began in the 1760s, and it led to advent of mechanization, steam power, and weaving loom. In the year 1870, mass production driven by electricity and assembly lines provided momentum for Industry 2.0. The third industrial revolution 3.0, beginning in the 1960s was the era of Automation and digital technologies – electronic chips, mainframes, PCs and the Internet. All this paved the way for the fourth industrial revolution (4.0) in the 21st century by the confluence of emerging technological breakthroughs, from artificial intelligence, 3D printing, nanotechnology, machine learning to quantum computing (Klaus Schwab, 2016).

The current study aims to deduce an approach for Predictive Maintenance using IoT sensors that will aid manufacturing industries to predict machine failures before asset breakdown occurs. This paper discusses a case study from the manufacturing sector where the solution is developed using Machine Learning's (ML) binary classification Support Vector Machine, Classification and Regression Trees (C&RT), and Boosted Classification Trees. These algorithms are discussed in Section 5.

The paper is organized as follows. Literature review on Predictive Maintenance (PdM) modeling and IIoT sensors are presented in Section 2. An Architecture for Maintenance modeling using IIoT sensors is proposed in Section 3. A brief overview of Problem definition, features, data set,

methodology, and machine learning models are presented in Section 4 of the case of manufacturing firm. Section 5 describes experiments, results, and discussion. Finally, conclusion and future scope are presented in Section 6.

2. Literature Review

IIoT (Industrial Internet of Things) has spread its applications into various verticals such as healthcare (Farahani & Firouzi et.al, 2018), manufacturing (Selcuk, 2016), oil and gas (Chiang & Zhang, 2016) and many more. With the use of Machine learning analytics, huge amounts of data generated from IIoT sensors fitted on manufacturing assets can be leveraged to detect asset failure in advance. The objective is to predict failures in advance, thereby increasing equipment uptime. The research on IIoT based Predictive maintenance is still in the preliminary stage. Earlier literature (Lee, Ghaffari & Elmeligy, 2011) was mainly centered around Condition Based Maintenance (CBM) Strategies (Shin, J.-H. & Jun, 2015), as described in section 1, which was a part of Industry 3.0 revolution.

Literature by (Lee, Ghaffari & Elmeligy, 2011) discusses smart maintenance systems that estimate the reliability of a system to detect any failures beforehand. It is based on maintenance techniques such as Preventive Maintenance (PM) and CBM strategies. In the research by (Selcuk, 2016), the latest trends, applications, and how PdM can help improve reliability, efficiency, and protect the environment. Work done in (Civerchia, Bocchino, Salvadori et.al, 2017) describes the system and component architecture of Industrial sensors in detail. Research by (Mourtzis, Vlachou & Milas, 2016) discusses the industrial application of IIoT wireless sensors networks how its adoption generates volumes of big data that needs to be further analysed. The above work focuses on PdM and IIoT applications, trends, CBM and Industrial IIoT sensor architecture. Implementation of IIoT sensor architecture using Machine learning algorithms has not yet been fully exploited.

With the advent of Industry 4.0, the goal of Predictive Maintenance systems is reaching Zero-Defect Manufacturing (Wang, 2016). The author defines PdM as a set of activities that can detect changes in a physical state of an asset (signs of possible equipment failure) so that the appropriate maintenance work for maximizing the service life of an asset can be done optimally. The author (Wang, 2016) propose an Intelligent Predictive Maintenance framework (IPdM) that gave an initial direction for the current research work. Overall, the goal is to create and deploy a model for prediction of machinery failure using data collected from industrial sensors. The next section discusses the architecture of the PdM model.

3. IoT based Predictive Maintenance Architecture

The study proposes an architecture for Industrial IoT based predictive maintenance (Fig. 1). IoT sensors are used to monitor and collect real-time data. The data collected is then modelled into PdM model using Statistical tools and software. The following are the components of PdM systems.

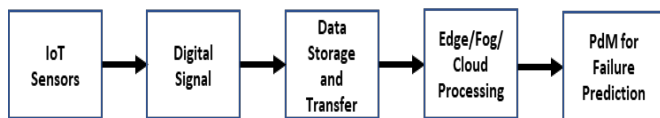


Fig. 1 - Architecture Explained

3.1. IoT Sensors

Sensors or Cyber-physical systems can interact and communicate with each other. Sensors capture raw data. Few Popular Industrial IoT Enabled Sensors measure Temperature, Humidity, Pressure, Current, Vibration, Air Quality, Footfall, Gas, and Weight. Another commonly used sensor is the Photoionization Sensor, which is used to measure levels of volatile inorganic compounds and gases. The ultrasonic sound sensor is used to translate high-frequency sounds due to leaks.

3.2. Digital Signal:

Data from sensors captured in the form of analog signals is converted into digital signals using Analog to digital converter. The digital signal is machine-readable and is used for further analysis.

3.3. Data Storage and Transfer:

Digital data stream generated is stored on organizations' local intranet or a secure centralized or remote cloud location in the form of a database file. IoT Sensors which are installed in remote locations without internet connectivity have inbuilt Random Access Memory (RAM) and Storage

space where data is stored. Data from IoT sensors is transmitted to a centralized cloud server or decentralized fog nodes via Wi-Fi, or in remote locations through Cellular networks or Bluetooth Low Energy (BLE) technology.

3.4. Edge/Fog/Cloud Computing:

IoT sensors like Arduino, Raspberry PI, ESP8266, ESP32 have inbuilt computing power. Based on the criticality of the asset, processing of PdM algorithms can be done at Edge, Fog, or Cloud. In Edge devices, the processing is done locally. It is used to provide near to real-time insights about Machine failure. In cloud computing, the server is located in a centralized location, having big data centers (Chiang & Zhang, 2016). Data generated from the IoT sensors in PdM applications are sent to Cloud for further processing. Processing at cloud has high latency. Fog Computing has distributed architecture where fog nodes are distributed in many locations, over large geographic areas (David Linthicum, 2016). It is the intermediate layer between Cloud and IoT Edge devices (Fig. 2) if the assets are separated by the large distance it is recommended to implement fog architecture over the cloud due to low latency and higher computing power (Atlam, Walters & Wills, 2018).

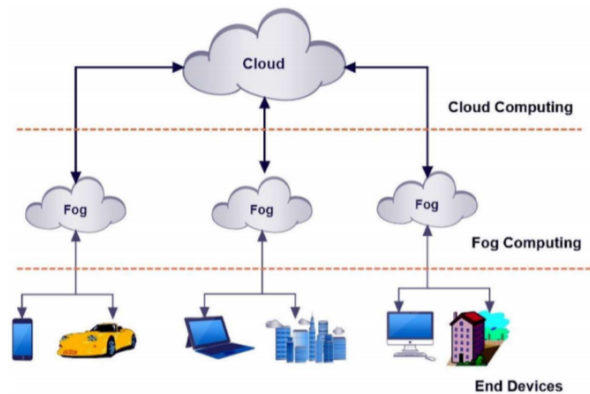


Fig. 2 - Fog Computing, an extension of Cloud computing

3.5. Predictive Maintenance:

After the data has been transferred to Fog/Cloud/Edge nodes or a centralized database, the next step is PdM algorithm design. The goal of predictive maintenance is to predict asset failure. ML algorithms (Supervised or unsupervised) are applied to model the data. The processing of the data from IoT sensors is done at fog/cloud or edge nodes. The algorithms process data sent from IoT sensors embedded in the asset and predict failures.

The steps discussed above are generalized and applies to all PdM systems in manufacturing. A case from the Auto Ancillary Company is discussed next.

4. Case Problem Definition

XYZ Pvt Ltd (name changed) is a manufacturer of automobile parts based in India. The manufacturing process at XYZ Company involves metallic products, plastic, and polymer products manufacturing. The assembly line of its metal manufacturing products consists of components like – motors, rotors, heat exchangers, robotic arms, etc. If any of these components get faulty, the entire assembly line needs to be stopped resulting in downtime. The automobile sector incurs huge losses due to downtime (Advanced Technology Services, 2006). XYZ Ltd. wants to predict machine failure in advance to take preventive measures before actual machine downtime occurs, thereby aiming to reduce its maintenance costs, increase productivity, improve product quality and save energy from smoother machines.

4.1. Data Collection from Critical Assets

XYZ Ltd., as a pilot planned to implement IoT based predictive maintenance on its heat exchanger unit whose function was to cool down extremely high-temperature synthetic fluids being discharged from the assembly line. The workstation had frequent issues of downtime on the clogging of its conduits due to chemical deposits and issues of heat leakage due to thermal cracks or perforations. The heat exchanger unit was critical in operation, and there were employee safety risks associated with its frequent failure. XYZ Ltd. wanted a cost-effective solution to monitor and predict failure. The workstation was connected with IoT sensors that are discussed next.

4.2. Dataset and Features

The dataset used for demonstration is publicly available as original data could not be shared. The sample size of the dataset is 944 observations and ten features. The dataset has been cleaned and does not contain any missing values or outliers. The dataset can be downloaded from <https://bit.ly/2nwwwiE>. Description of features in the dataset is as follows:

- footfall: The total number of workers in the shop floor at a particular instance of time
- tempMode: temperature mode (1-7), where one is minimum, and seven is maximum
- AQ: Air Quality sensor, with modes between 1 to 7, where modes 1-5 are acceptable levels of air quality and mode 6 -7 are considered not acceptable
- : Ultrasound Sensor, used to detect high or low-frequency sounds due to leakage.
- CS: Current Sensor to monitor the current flow. Its level is between 1-7, where 1 in minimal level and 7 is maximum current flow
- VOC: (PhotoIonization Detector) sensors used for detecting low concentrations of Volatile Organic Compound to detect clogging in conduits of heat exchanger unit.

- RP: Outside room Pressure.
- IP: Inside Pressure of machine conduits for leak detection.
- Temperature: actual temperature of machinery
- Output: The target variable predicting whether the equipment is running or failed. It is a binary coded variable, where 0 depicts running, and 1 represents failure.

4.3. Methodology

The approach is spread across six phases. (1) In the first phase, critical assets of the organization are identified. These include assets that are critical to the organization, have frequent failure rates, and whose maintenance involves high costs. (2) Data is collected from IoT sensors installed on machines, as discussed in Section 3, is carried out in the second phase. This data is stored in a database or cloud. (3) In the third phase, the raw data is pre-processed. Much of raw data in databases may contain missing values, outliers, obsolete or redundant data and inconsistent values (Larose & Larose, 2015). The data need to be prepared, so that clean data generates higher reliability and accuracy of prediction. (4) In the next phase, data were split into training and test data. Supervised machine learning algorithms were applied to create the PdM model. The model is trained using training data and model performance is checked on the test data. (5) In the fifth phase, the output is analyzed and is evaluated based on the accuracy of prediction of the ML algorithms. This is done using Receiver Operating Curve (ROC) metrics, Gains charts, visualizations such as Key Performance Indicator charts, dashboards, confusion matrix, misclassification error rates, and there are several other performance metrics, which can be used as required. (6) In the final phase, the model is deployed to the enterprise. Deployment of the model is done in the production stage. Machine learning tools like Microsoft Azure ML Studio (Software), Tibco Statistica (Software), SAS Visual Data Mining and Machine learning (Software), Google AI platform (Software) and open-source software like R and Python can be used to model the data and trigger alarms on a prediction of failure.

5. Model Building

TIBCO Statistica version 13.5 (Software) is used to model the IoT sensor data, using supervised learning algorithms. Statistica is a powerful predictive analytics software that allows end-users to build models that can be packaged and published by businesses. TIBCO Statistica provides a visual drag and drop mode of interaction for modelling known as a workspace (Fig. 3).

5.1. Steps for Model Building

The model (Fig. 3) that is deployed as a pilot for XYZ Ltd. A brief description of each of the vital components of the workflow is as follows.

- Input dataset into the TIBCO Statistica workspace.

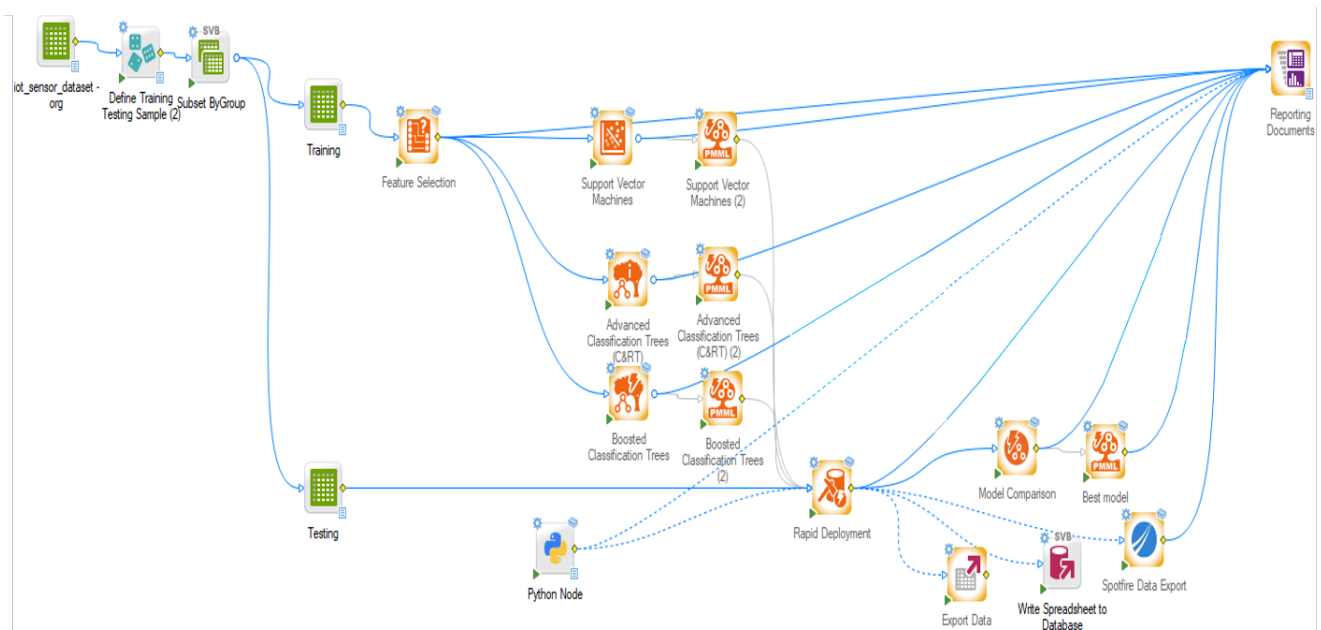


Figure 3: Model Building on Tibco Statistica

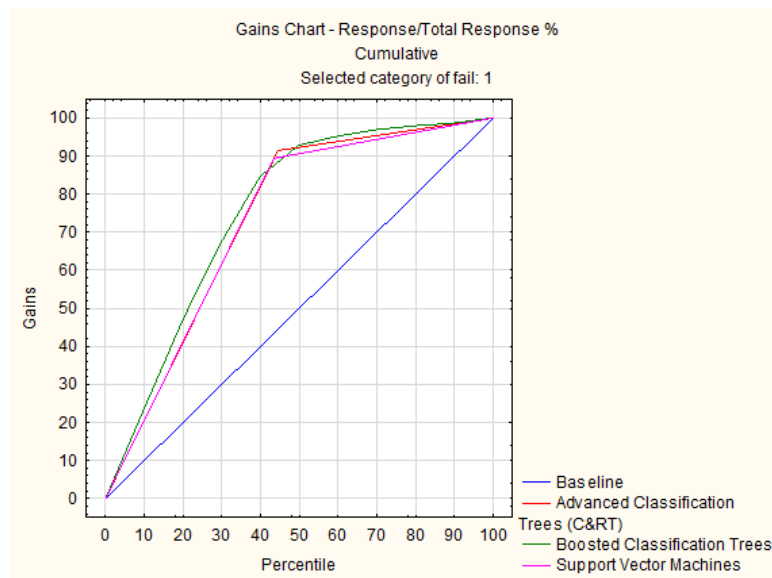


Figure 4: Gains Chart

- *Training Testing Sample*: Data was split 75:25 ratio as Training and Test sample
 - *Predictor screening*: An option to select dependent/independent variables to be used for analysis. Variable 'Output' was set as the categorical dependent variable. Rest of the variables in our dataset like 'Footfall,' 'tempMode,' 'AQ,' 'USS,' 'CS,' 'VOC,' 'RP,' 'IP' and 'Temperature' were set as continuous predictors
 - *Algorithm*: Users can select various Data mining and Machine learning algorithms to model data. We have used binary classification algorithms to model our data. A brief detail of classification algorithms used for our PdM model is
- Support Vector Machine (SVM): SVM represents a powerful technique for the general classification, regression and outliers detection with an intuitive model representation (Suykens & Vandewalle, 1999).
- Random Forest: An Ensemble method for classification and regression (Liaw & Wiener, 2002).
- Advanced Classification Trees (C&RT): Advanced decision trees for classification and regression (Lemon, Roy, et.al, 2003).
- Each algorithm generates a PMML (Predictive Model Markup Language) script file. PMML is an open standard to represent predictive models.
- *Rapid Deployment*: This node applies PMML codes to testing data and outputs predicted values and model comparison metrics. Rapid deployment node evaluates methods that measure model performance. Deployment of a model in Production is done by publishing PMML codes to an enterprise by a web service.
 - *Reporting*: Output model summary report and prediction accuracy. The algorithms are compared using the same training and testing samples. The output data can be saved and exported to Tibco Spotfire (a tool for data modelling and visualization) or database export or simple export into static Excel/CSV files.

5.2. Results and Discussion

In this section, we analyse the predictive maintenance model by comparing the results of three machine learning models on the training-testing dataset. A comparison and summary of error rates on testing data [Table 1] show that Boosted Classification Trees output minimum error in predicting machine failure. The error rate is calculated based on a voted prediction of the output of all the three algorithms. If two out of three algorithms predict Machine failure and the third one does not predict machine failure, based on a maximum count of votes, the output would be that Machine failure would occur.

The next important metrics to analyse are Precision, recall and F1 score. Precision is the ratio of correctly classified observations. A recall is the ratio of correctly predicted positive events. A recall is also known as sensitivity (Ruiter, 2015). Recall metric signifies the ratio of correctly predicted positive events. F1 score is a harmonic average of Precision and recall. F1 metric is used when class distribution is uneven. In the IoT sensor dataset, the ratio of 'Fail' and 'Pass' classes are not equal. A comparison of Precision, Recall and F1 metric is shown [Table 1].

Table 1 - Summary of Model Performance Metrics on Testing data

Metric	C&RT	Boosted Classification Trees	SVM
Precision	0.891	0.899	0.893
Recall	0.914	0.908	0.894
F1 Score	0.903	0.903	0.893
Error Rate	0.099	0.097	0.106

Fig. 4. shows the Gains chart. The chart interprets the utility that one can expect by using C&RT, SVM and Boosted tree algorithms, as compared to using baseline information only or in other words the gains chart represents the ratio of results obtained with and without application of any models. The curve with maximum area under curve represents optimal model. As can be observed, Boosted Classification tree algorithm has maximum area under curve.

PdM architectural framework and steps proposed by (Wang, 2016) was implemented on our model using higher accuracy obtained by voted average of three machine learning algorithms. The model was evaluated based on Misclassification error rate, Precision, Recall, F1 score, and Gains chart. It is worthwhile to note that in case of PdM where cost of a false negative (when the production line failure occurs and the model is unable to predict it) is higher than false positive and hence it is best to focus on Recall metric to evaluate the model. The results show that the model can predict failure with a maximum accuracy rate of 90.18% on the IoT sensor dataset and using the C&RT algorithm model has the best recall.

The next step is to deploy the model to production via a web service using PMML codes that were generated by Tibco Statistica. Further setup can be done to trigger alarm (through email, SMS notification) on machine fault prediction and monitor system status and production line data using real time dashboards.

6. Conclusion

The proposed work discusses IoT based predictive maintenance and implements it using a case from the manufacturing industry. The research paper initially discusses Industry 4.0 and an interconnected IoT based Sensor ecosystem that can share and exchange information amongst each other. Data collected from sensors can be leveraged by manufacturing organizations to predict machine failure.

Opportunities in IoT and implementation of Predictive maintenance in industries will be huge. The ability to predict failure in advance will help organizations increase productivity, reduce equipment costs, improve working conditions and work environment safety, better product quality, reduced waste in terms of consumables and energy savings. On average, the industry has seen a 70%-75% reduction in machine downtime (Selcuk, 2016).

The case discusses a pilot implication of PdM for a specific asset in

the assembly line of the manufacturing industry. A model is created using advanced analytics software, which tends to simplify the process.

A successful implementation of Industrial PdM would involve any organisation to scale across multiple assets. A model having thousands of IoT sensors would generate huge volumes of data. Such huge volumes of data need to be stored and analysed using Big data analytics. Apart from this future scope for our research would also involves implementing a Prescriptive maintenance-based solution for predicting failure and what action need to be taken to resolve it.

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- Supplement Information: The data set used in the study can be downloaded from <https://bit.ly/2nwwiE>. The feature names have been modified for better understanding of the case. Data Source: Sapiro, Virginia, Steven J. Rosenstone, Donald R. Kinder, Warren E. Miller, and the National Election Studies. *American National Election Studies, 1992-1997: Combined File* [Computer file]. 2nd ICPSR version. Ann Arbor, MI: University of Michigan, Center for Political Studies [producer], 1999.